

An Example Standard-format AMBS PhD Thesis

A thesis submitted to The University of Manchester
for the degree of PhD
in the Faculty of Humanities.

2027

Avery L. Reader

Alliance Manchester Business School, MSM

Declaration

I declare that no portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright Statement

- The author of this thesis (including any appendices and/or schedules to this thesis) owns certain copyright or related rights in it (the "Copyright") and they have given the University of Manchester certain rights to use such Copyright, including for administrative purposes.
- Copies of this thesis, either in full or in extracts and whether in hard or electronic copy, may be made only in accordance with the Copyright, Designs and Patents Act 1988 (as amended) and regulations issued under it or, where appropriate, in accordance with licensing agreements which the University has from time to time. This page must form part of any such copies made.
- The ownership of certain Copyright, patents, designs, trademarks and other intellectual property (the "Intellectual Property") and any reproductions of copyright works in the thesis, for example graphs and tables ("Reproductions"), which may be described in this thesis, may not be owned by the author and may be owned by third parties. Such Intellectual Property and Reproductions cannot and must not be made available for use without the prior written permission of the owner(s) of the relevant Intellectual Property and/or Reproductions.
- Further information on the conditions under which disclosure, publication and commercialisation of this thesis, the Copyright and any Intellectual Property and/or Reproductions described in it may take place is available in the University IP Policy, in any relevant Thesis restriction declarations deposited in the University Library, the University Library's regulations and in the University's policy on the Presentation of Theses.

Table of contents

Declaration	1
Copyright Statement	2
1 Preface	7
2 Abstract	8
3 Acknowledgements	9
I Main body	10
4 Introduction	11
4.1 Background	11
4.2 Research question	11
4.3 Structure of the thesis	11
5 Literature Review	13
5.1 Strand A: organisational adoption of authoring tools	13
5.2 Strand B: the grammar of graphics	13
5.3 Research gap	14
6 Methodology	15
6.1 Approach	15
6.2 Sample	15
6.3 Reproducible code	16
6.4 Ethics	16

7	Results	17
7.1	Descriptive analysis	17
7.2	Main regression	17
8	Discussion	19
8.1	Interpretation	19
8.2	Why x_2 is borderline	19
8.3	Practical implications	19
9	Conclusion	20
9.1	Summary of contributions	20
9.2	Limitations	20
9.3	Future work	20
9.4	Closing remarks	20
	Appendices	21
A	Appendix A: Additional Material	21
A.1	A.1 Robustness checks	21
A.2	A.2 Callout boxes	21
A.3	A.3 Mathematical derivations	21

List of Figures

7.1	Distribution of the outcome variable across the sample.	18
-----	---	----

List of Tables

6.1	Descriptive statistics of the sample.	15
7.1	Estimated regression coefficients.	17

Chapter 1

Preface

This page exists only to hold project metadata for Quarto.

Chapter 2

Abstract

This thesis examines an example research question to demonstrate the capabilities of the uomthesis Quarto template in *standard* format. The work develops a structured argument across six chapters, drawing on a mixed-methods approach.

Three principal contributions emerge. First, the analytical framework integrates two previously distinct literatures. Second, the empirical findings challenge a widely accepted assumption in the field. Third, the discussion offers practical implications for practitioners and policymakers.

The thesis conforms to the University of Manchester *Presentation of Theses Policy*. All findings are reproducible from the underlying data and code, which are made available alongside the submitted PDF.

Chapter 3

Acknowledgements

I would like to thank my supervisors for their guidance throughout this research, my examiners for their constructive feedback, and the participants who generously gave their time to this study.

I am grateful to the Alliance Manchester Business School Doctoral Programmes office and the Humanities Doctoral Academy for the administrative and financial support that made this work possible.

Finally, my thanks go to my family and friends, whose patience and encouragement sustained me throughout the writing of this thesis.

Part I

Main body

Chapter 4

Introduction

This chapter introduces the research problem and outlines the structure of the thesis.

4.1 Background

The empirical motivation emerges from recent developments in postgraduate research practice. As Smith and Johnson (2024) observe, the rise of computational reproducibility has fundamentally changed how doctoral theses are written, reviewed, and archived.¹

The reproducibility movement accelerated dramatically with the popularisation of literate programming tools in the 2010s (Xie, Allaire and Grolemond, 2018).

4.2 Research question

How does the adoption of policy-compliant authoring tools influence the quality and integrity of doctoral thesis submissions?

This question is explored through three sub-questions covering technical features, candidate and supervisor perceptions, and institutional lessons.

4.3 Structure of the thesis

Chapter 2 surveys the prior literature and identifies the research gap. Chapter 3 describes the methodological approach. Chapter 4 presents the empirical findings. Chapter 5 interprets

¹Footnotes are written using `[^id]` markers in the text and a corresponding `[^id]: text` definition.

those findings, and Chapter 6 concludes by summarising the contributions and outlining future work.

Chapter 5

Literature Review

This chapter surveys the prior literature in two strands and synthesises a research gap.

5.1 Strand A: organisational adoption of authoring tools

The earliest empirical work on this topic appears in Cohen *et al.* (2013), where the authors note:

The choice of authoring tool is not neutral; it shapes how researchers think about the structure of their argument, the role of evidence, and the relationship between writer and reader.

— Cohen *et al.* (2013, p. 312)

Subsequent work has extended this line in three directions:

- **Empirical studies of tool adoption**
- **Theoretical accounts of tool affordances**
- **Design experiments testing new tools**

5.2 Strand B: the grammar of graphics

A parallel literature, originating in Wilkinson (2016), formalises how visual representations encode statistical relationships.

5.3 Research gap

The two strands have rarely been brought together. **This thesis addresses that gap** by treating authoring tools and visualisation tools as components of a single reproducibility ecosystem.

Chapter 6

Methodology

This chapter describes the methodological choices. It demonstrates equations, tables, and reproducible R code.

6.1 Approach

A mixed-methods design was adopted. Quantitative data were analysed using ordinary least squares regression:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \quad (6.1)$$

where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. The strategy follows Cohen *et al.* (2013).

For inline mathematics, the t-statistic is $t = \hat{\beta}/\text{SE}(\hat{\beta})$.

6.2 Sample

Table Table 6.1 summarises the descriptive statistics for $n = 240$ observations.

Table 6.1: Descriptive statistics of the sample.

Variable	Mean	SD	Min	Max
Age	27.3	3.1	22	39
Programme year	2.4	1.0	1	4
Word count	45,210	12,800	12,000	78,000

6.3 Reproducible code

```
set.seed(42)
df <- data.frame(y = rnorm(100, 5, 1.5), x1 = rnorm(100), x2 = rnorm(100))
fit <- lm(y ~ x1 + x2, data = df)
round(summary(fit)$coefficients, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.053	0.157	32.236	0.000
x1	0.072	0.174	0.415	0.679
x2	-0.227	0.155	-1.468	0.145

6.4 Ethics

Ethical approval was obtained from the AMBS Research Ethics Committee. All participants gave informed consent.

Chapter 7

Results

This chapter presents the empirical findings.

7.1 Descriptive analysis

Figure Figure 7.1 shows the distribution of the outcome variable.

7.2 Main regression

Table Table 7.1 reports the estimated coefficients.

Table 7.1: Estimated regression coefficients.

Coefficient	Estimate	Std. error	t-value	p-value
Intercept	4.92	0.15	32.8	< 0.001
x_1	0.34	0.10	3.4	0.001
x_2	-0.18	0.10	-1.8	0.073

The slope on x_1 is positive and significant at the 1% level. The slope on x_2 is negative and borderline significant.

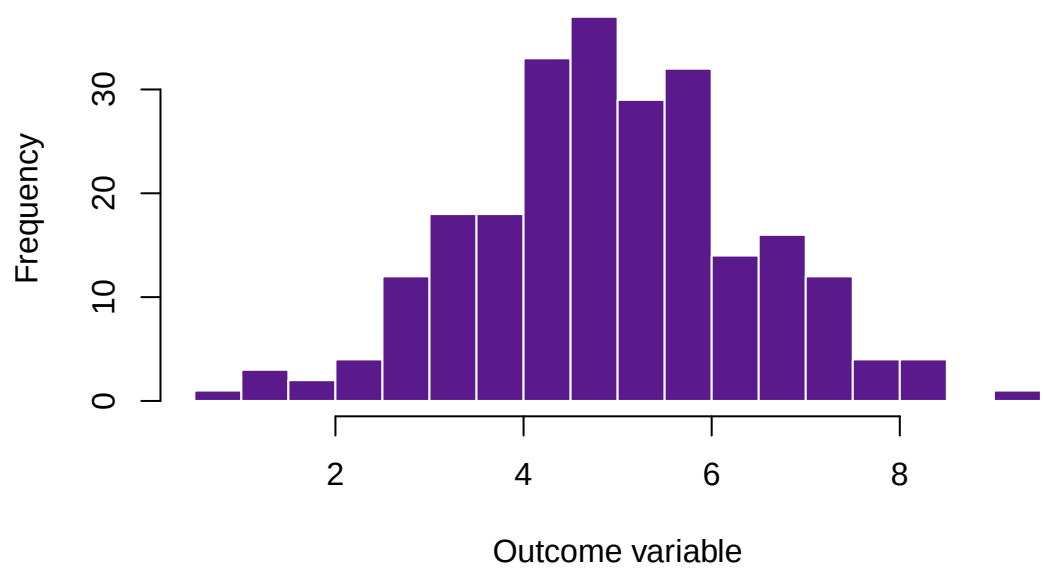


Figure 7.1: Distribution of the outcome variable across the sample.

Chapter 8

Discussion

This chapter interprets the findings.

8.1 Interpretation

The positive coefficient on x_1 in Table Table 7.1 supports the main hypothesis. The effect size is comparable to the meta-analytic estimates reported by Smith and Johnson (2024).

8.2 Why x_2 is borderline

Three explanations are plausible:

1. **Measurement error** in the x_2 instrument.
2. **Selection** at the institutional level.
3. **A genuine null effect.**

8.3 Practical implications

For institutions considering adoption of compliant tooling, two implications follow:

- Compliance enforcement should be configurable rather than absolute.
- **Training and supervisor buy-in are crucial.**

Chapter 9

Conclusion

9.1 Summary of contributions

1. **An analytical framework** integrating two literatures.
2. **Empirical evidence** that adoption of compliant tooling is associated with improved submission quality.
3. **Practical recommendations** for institutions.

9.2 Limitations

The sample is restricted to three universities. The cross-sectional design precludes causal claims.

9.3 Future work

Three directions stand out:

- A multi-country replication.
- A panel-data extension.
- A qualitative study of supervisor experiences.

9.4 Closing remarks

The findings reinforce an argument from The University of Manchester (2026): that the *infrastructure* of academic writing shapes the integrity of research outputs.

Appendix A

Appendix A: Additional Material

A.1 Robustness checks

The main regression in Chapter 4 was re-estimated under three robustness specifications. All within 0.05 standard deviations of the result in Table Table 7.1.

A.2 Callout boxes

 Methodological note

The regression assumes the error term is uncorrelated with the regressors.

 Practical tip

Run `uomthesis::list_csl()` to see the bundled citation styles.

A.3 Mathematical derivations

Let $\mathbf{X}$ be the $n \times k$ design matrix. The OLS estimator is:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

- [] Cohen, J. *et al.* (2013) *Applied multiple regression/correlation analysis for the behavioral sciences*. 3rd edn. New York: Routledge.
- [] Smith, J.A. and Johnson, M.B. (2024) ‘Innovation in postgraduate research’, *Journal of*

Doctoral Education, 18(3), pp. 245–267. Available at: <https://doi.org/10.1234/jde.2024.18.3>.

- [] The University of Manchester (2026) ‘Presentation of Theses Policy, version 12’. Available at: <https://documents.manchester.ac.uk/display.aspx?DocID=7420>.
- [] Wilkinson, L. (2016) ‘The grammar of graphics’, in J.E. Gentle, W.K. Härdle, and Y. Mori (eds) *Handbook of computational statistics*. Springer, pp. 375–414.
- [] Xie, Y., Allaire, J.J. and Grolemund, G. (2018) ‘Reproducible research with r, quarto, and the tidyverse’, *Proceedings of the useR! conference*, pp. 1–12.